



$$\left| \mathcal{A}_{3,\frac{1}{2},1,\frac{1}{4}}^z(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2$$

$$\left| -\frac{\sin \theta_q}{2} \cos \left[\frac{1}{2} \left(\phi_{q_{\parallel}} - \frac{\pi}{2} + \alpha \right) \right] G_{|-1|,|2|}(q_{\parallel}) \right|^2$$

$$\left| \frac{\sin \theta_q}{2} i \sin \left[\frac{1}{2} \left(\Delta J \phi_{q_{\parallel}} - \frac{\pi}{2} + \alpha \right) \right] H_{|-1|,|2|}(q_{\parallel}) \right|^2$$

